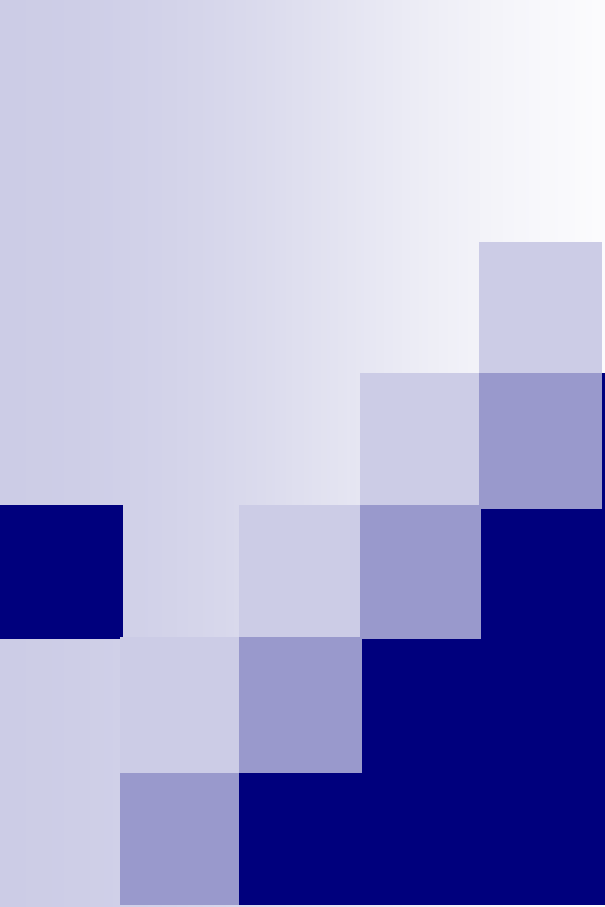
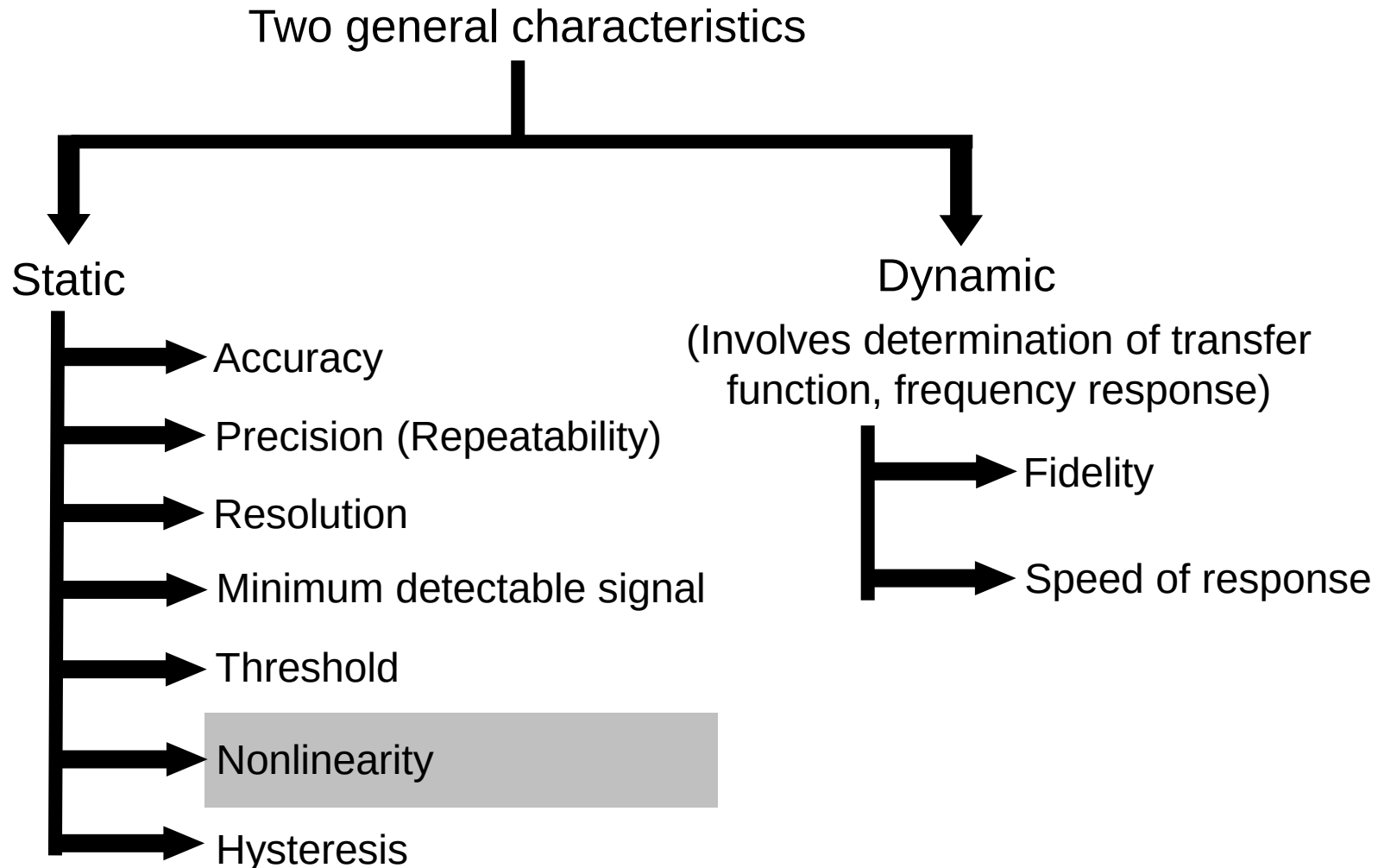




Design, Development and Performance Evaluation of Intelligent Sensors with Neural Networks

Sensor Characteristics

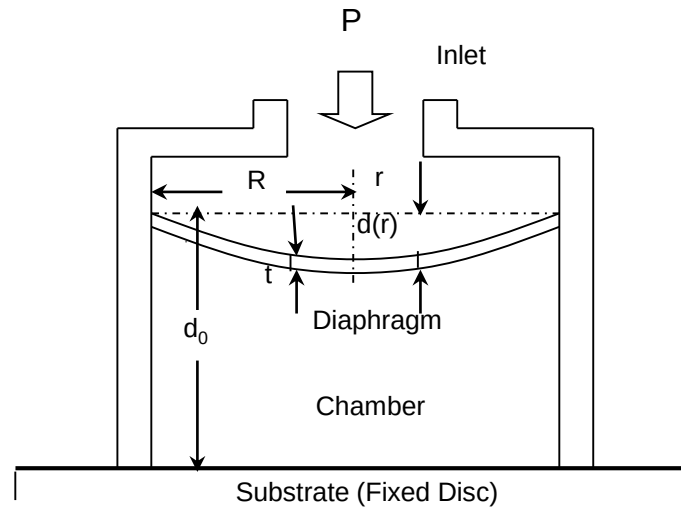


- Sensors exhibit inherent **nonlinear input-output characteristics**.
- **Nonlinearity** also creeps in due to change in environmental conditions such as **temperature** and **humidity**. **Aging** also introduces nonlinearity.
- Due to such nonlinearities, the range of the sensor **gets restricted** and hence the accuracy of measurement is affected if the **full range** of the instrument is used.

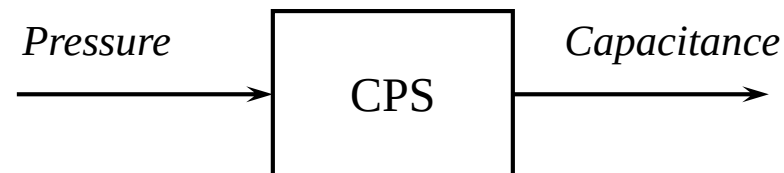
Objectives

- To study the nonlinearity problem of these devices (**CPS, LVDT and control valve actuator**) in-depth and suggest novel methods of circumventing the effect of this problem.
- The **adaptive** or **Artificial Neural Network (ANN)** based techniques are better candidates for such problems.

Capacitive Pressure Sensor



Typical structure of a CPS



The output capacitance,

$$C(x) = C_0 \frac{1 - \alpha x}{1 - x} = C_0 + \Delta C(x) \quad (A)$$

Where,

$$\Delta C(x) = C_0 \frac{1 - \alpha}{1 - x} x$$

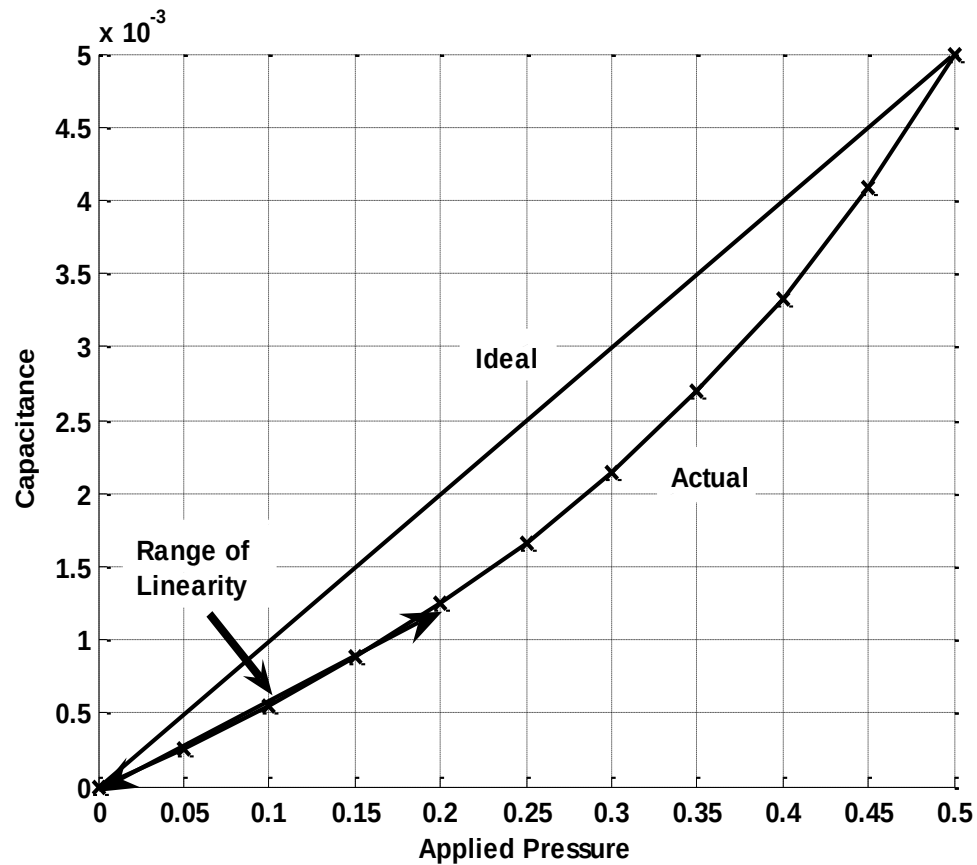
$$C_0 = \frac{\pi R^2 \varepsilon_0 \varepsilon_r}{d_0} = \text{Offset capacitance at } P = 0$$

$$x = \frac{P}{P_{\max}} (\leq 1) \text{ and } \alpha = \text{sensitivity parameter.}$$

ε_0 = permittivity of free space (vacuum),

ε_r = relative permittivity or dielectric constant of the insulating material.

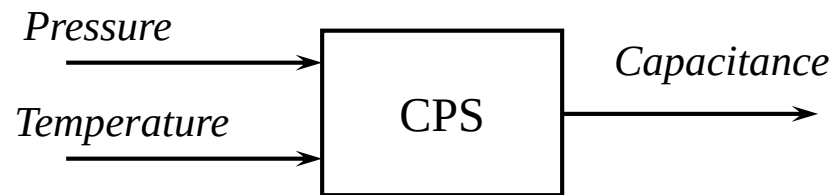
Hence by (A)



Range of linear region of the CPS

Since, nonlinearity varies with the environmental changes such as **temperature** and **humidity**.

Considering the temperature as another input parameter,



$$C(P, T) = C_0 (1 + \beta_1 \cdot T_N) + \Delta C(P, T_0) (1 + \beta_2 \cdot T_N)$$

$$C_N = C(P, T) / C_0 = (1 + \beta_1 \cdot T_N) + P_N \cdot \frac{1 - \alpha}{1 - P_N} (1 + \beta_2 \cdot T_N)$$

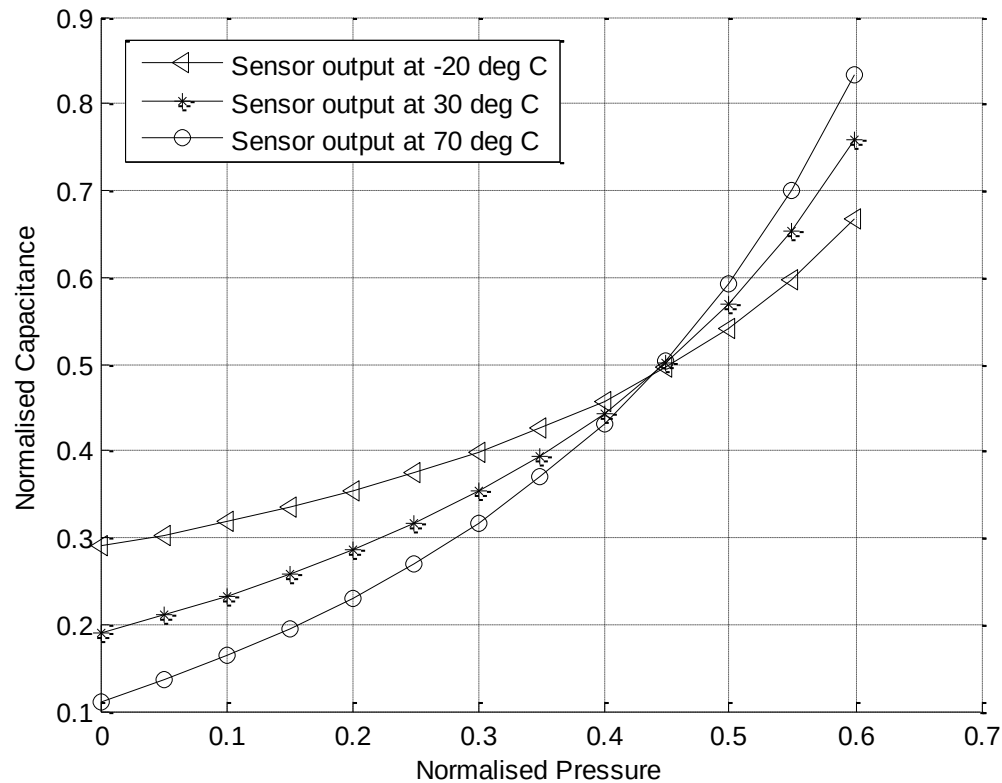
Where,

$$T_N = \text{normalized temperature} = \frac{T - T_0}{T_{\max}}, \quad T_0 = \text{reference temperature}$$

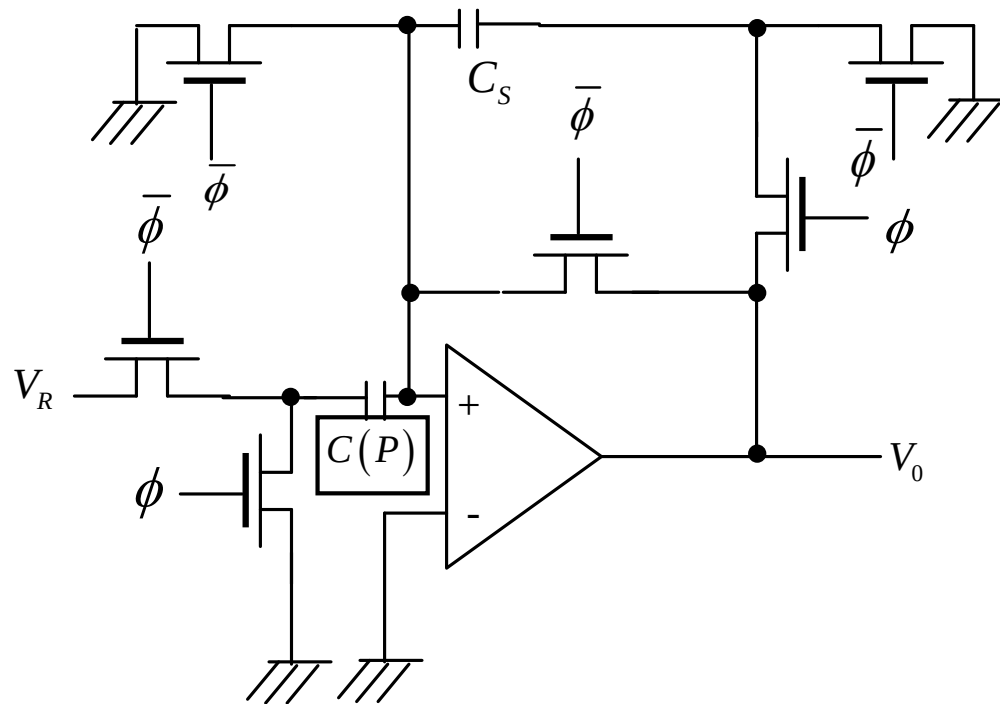
$$P_N = \text{normalized pressure} = P / P_{\max}$$

β_1, β_2 = coefficients, determines the influence of the temperature on the sensor characteristics.

The sensor characteristics at different temperatures,
At $T = -20^{\circ}\text{C}$, 30°C and 70°C



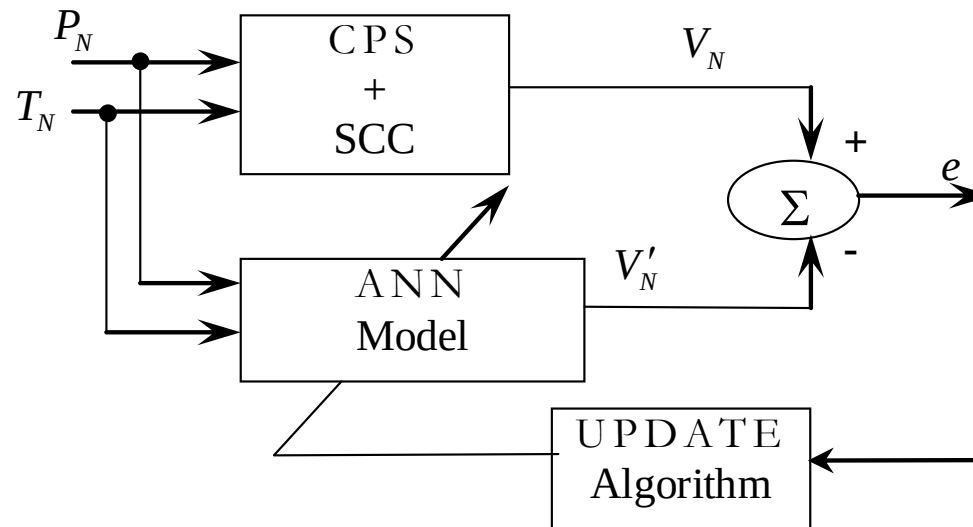
Switched Capacitor Circuit (SCC)



The SCC output provides a *voltage signal* proportional to the *capacitance change* in the CPS due to applied pressure.

Direct Modeling

- Importance of the direct modeling is to obtain an ANN model of the CPS in such a way that the outputs of the CPS and the ANN match closely.
- This is used for **fault detection of the sensor**.
- It is analogous to that of the system identification problem in control system.



The direct model is designed by applying different ANN techniques, such as

- Multilayer Perceptron (MLP)
- Functional link Artificial Neural Network (FLANN)
- Radial basis function (RBF) based ANN

MLP based Direct Modeling

A two-layer MLP with 3-5-1 structure is chosen for direct modeling of the CPS. (3, 5, and 1 denote the number of nodes including the bias units in the input layer, the first layer, i.e. the hidden layer and the output layer of the MLP, respectively).

The hidden and the output layer consists *sigmoid* type of activation function. The weights of the MLP are updated by the BP algorithm.

The learning rate and the momentum parameter are 0.5 and 1 respectively.

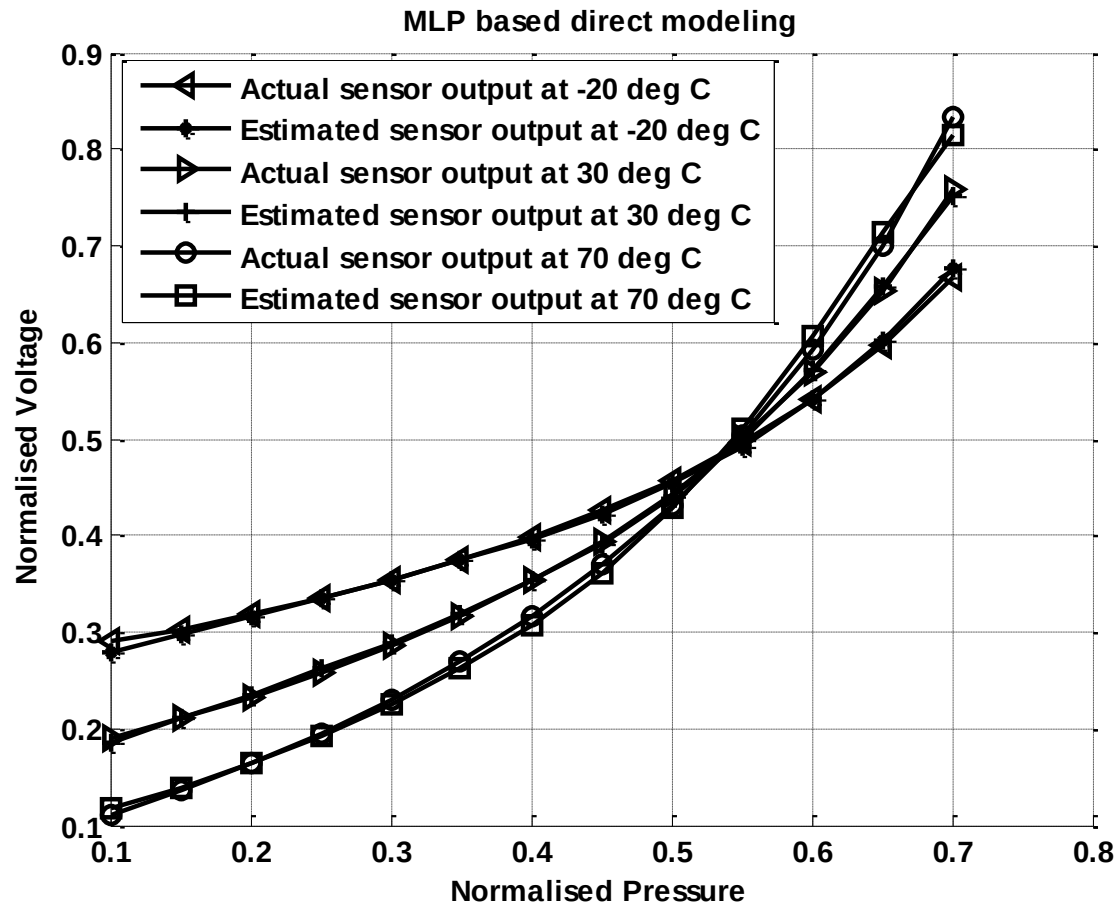
The parameters of the CPS are

$P_N = 0.1$ to 0.7 with a step of 0.05 ,

$T = -20^{\circ}\text{C}$ to 70°C with an interval of 10°C ,

$T_0 = 25^{\circ}\text{C}$, $\alpha = 0.7$, $\beta_1 = -2.0 \times 10^{-3}$ and $\beta_2 = 7.0 \times 10^{-3}$.

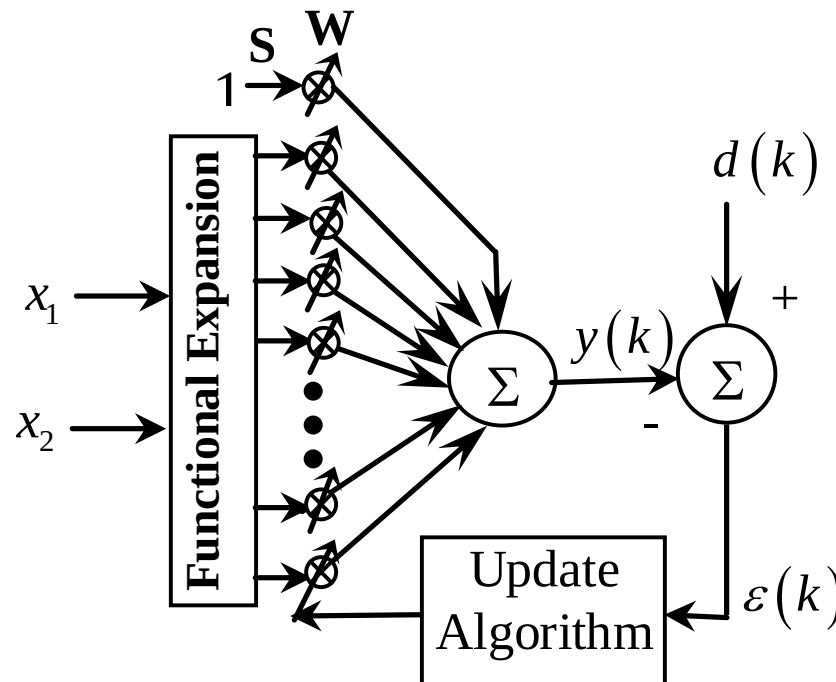
Simulation result by MLP



FLANN based Direct Modeling

It is a single layered network. Here, the input elements get functionally expanded nonlinearly, then weighted and summed up to yield the output.

The types of nonlinear expansion usually employed are trigonometric, power series, and tensor or outer product.

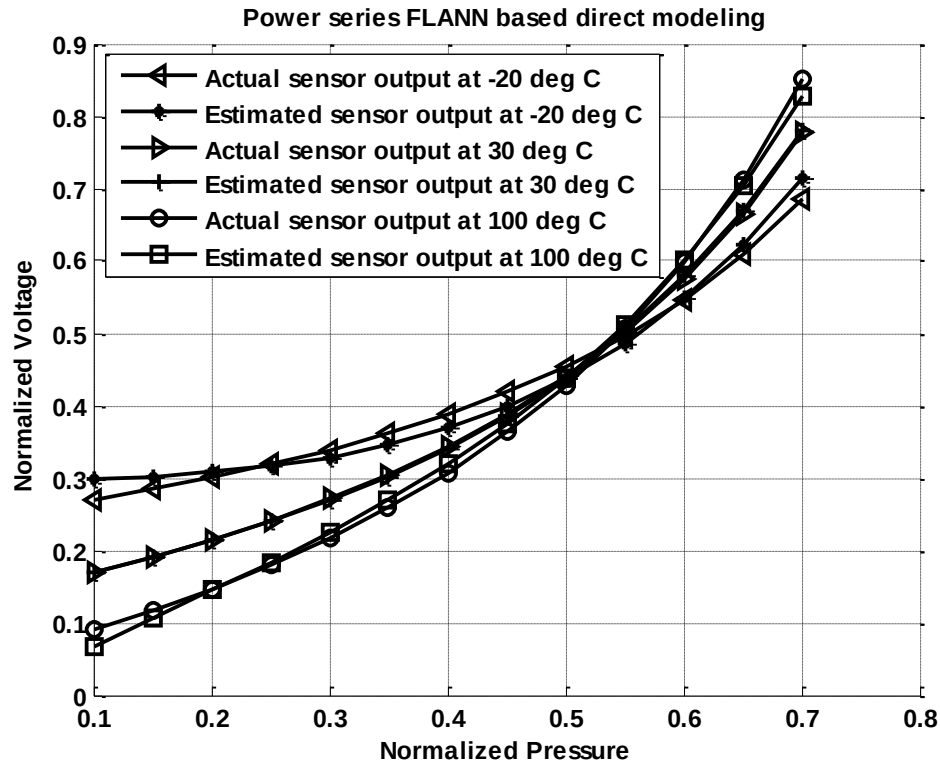


Simulation result of the power series FLANN

Temperature range, $T = -20^{\circ}\text{C}$ to 100°C with an interval of 10°C

No. of functional expansion = 5, (of each input)

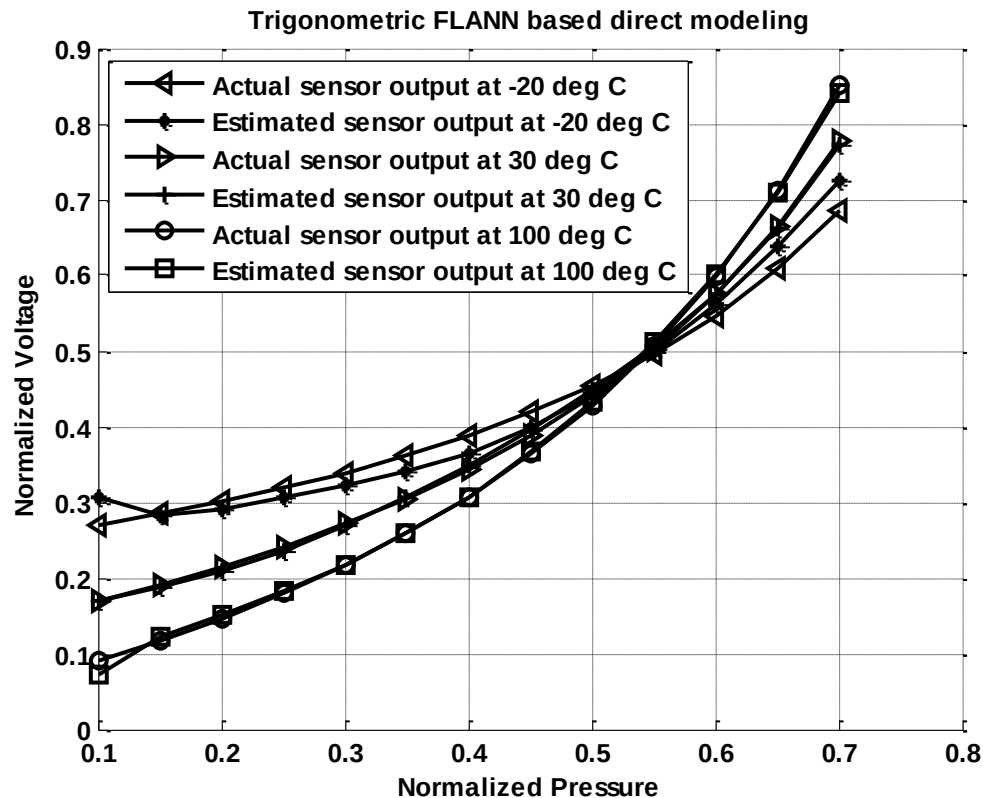
Learning rate of the update algorithm = 0.035.



Simulation result of the trigonometric FLANN

No. of $\sin(\cdot)$ expansion = 5, No. of $\cos(\cdot)$ expansion = 5, (of each input)

Learning rate of the update algorithm = 0.035.



RBFNN based Direct Modeling

Unlike the MLP, it is having one hidden layer. Each node of the hidden layer contains the radial basis function. i.e.

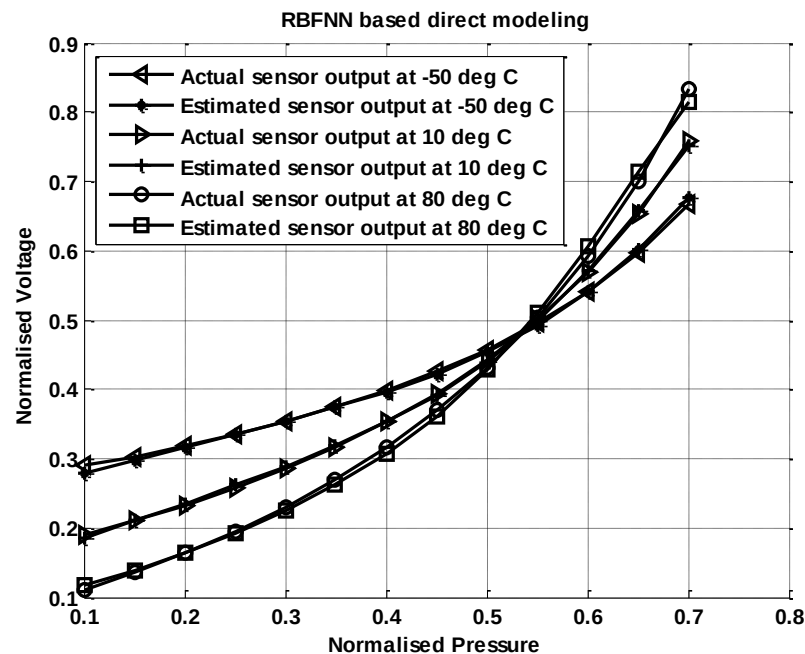
$$\phi_k(x_n) = \exp\left(-\frac{1}{\sigma_k^2} \|x_n - \mu_k\|^2\right)$$

2-6-1 RBFNN is taken for simulation. The learning rate of the updation algorithm is chosen as 0.05

Simulation result

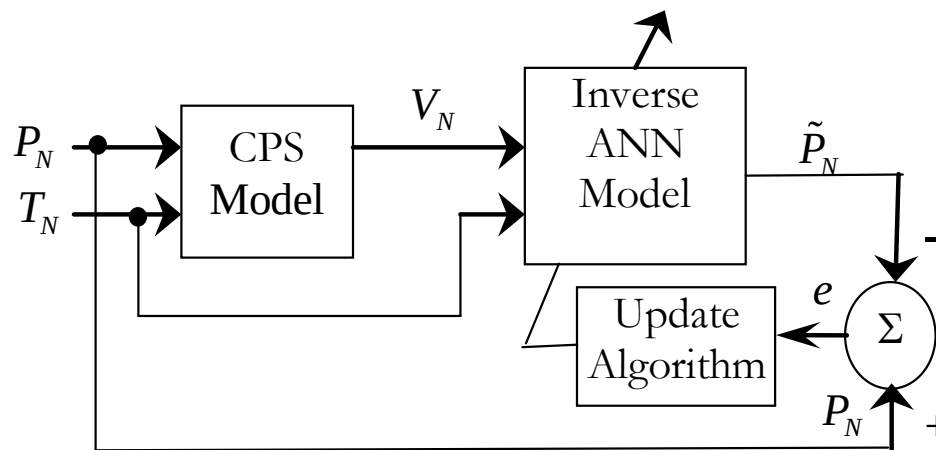
Temperature range,

$T = -50^{\circ}\text{C}$ to 80°C with an interval of 10°C



Inverse Modeling

- Inverse modeling of the CPS is designed to **estimate the applied input pressure**.
- This is analogous to the channel equalization scheme used in a digital communication receiver to cancel the adverse effects of the channel.
- This model is used in cascade with the sensor to compensate the nonlinearity effect.

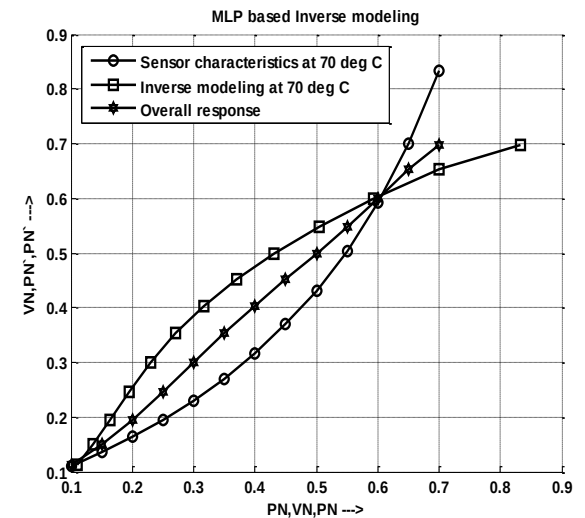
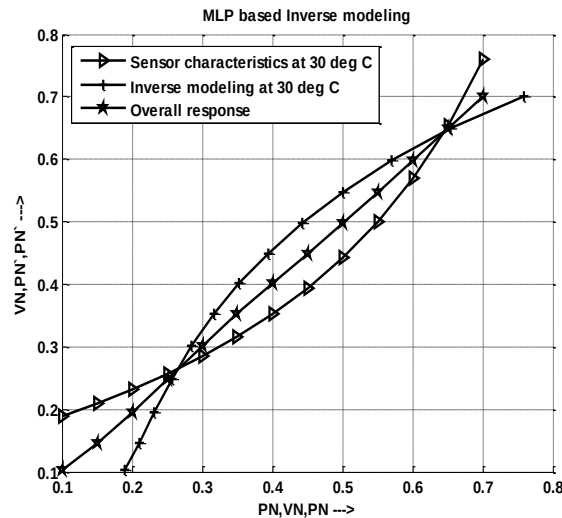
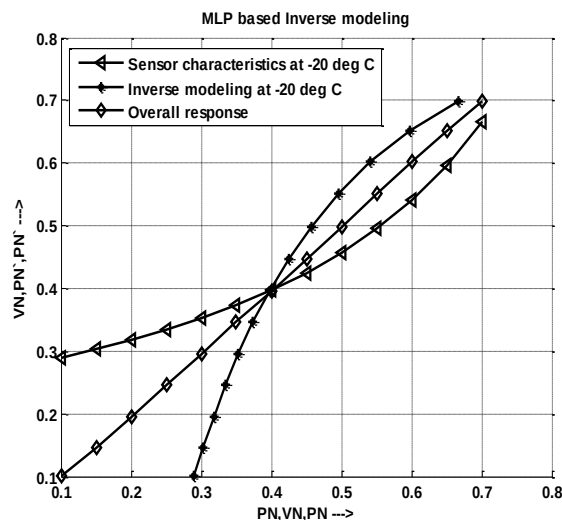


MLP based Inverse Modeling

In the simulation, the same MLP network is chosen for inverse modeling.

The learning rate and the momentum parameter are 0.5 and 1 respectively.

Figures are plotted at $T = -20^{\circ}\text{C}$, 30°C and 70°C .



Axes Parameters:

P_N = Actual input pressure.

V_N = Output of the CPS+SCC.

P_N' = Estimated pressure.

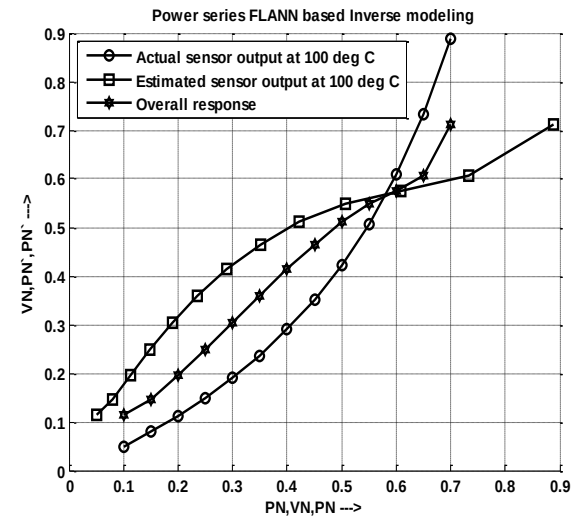
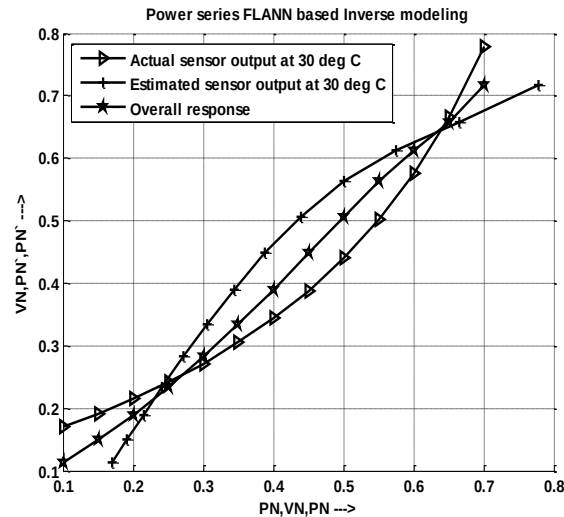
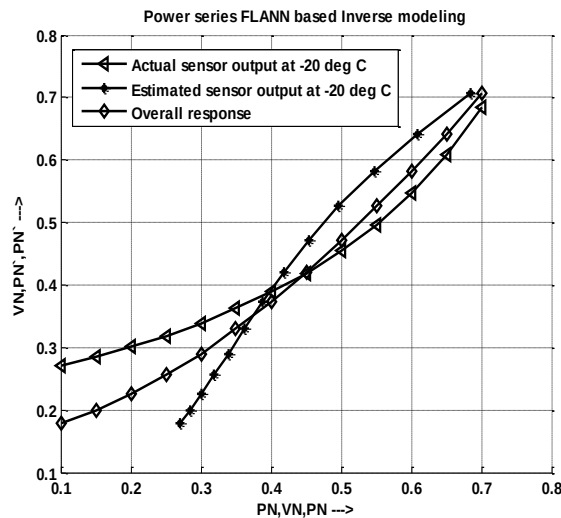
FLANN based Inverse Modeling

Simulation result of power series FLANN

No. of functional expansion = 12, (of each input)

Learning rate of the update algorithm = 0.035.

At different temperatures $T = -20^{\circ}\text{C}$, 30°C and 100°C .

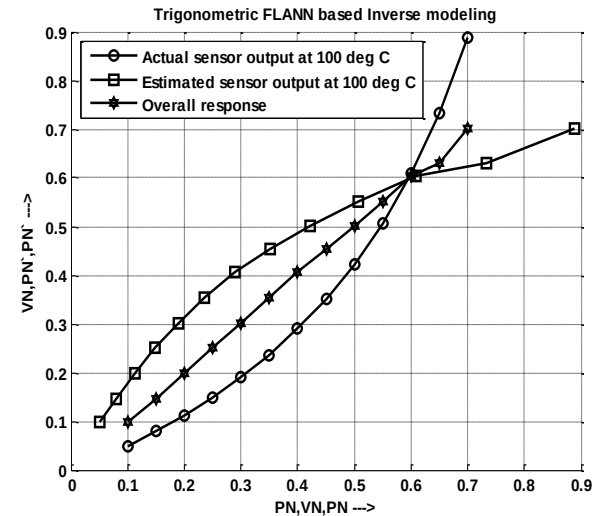
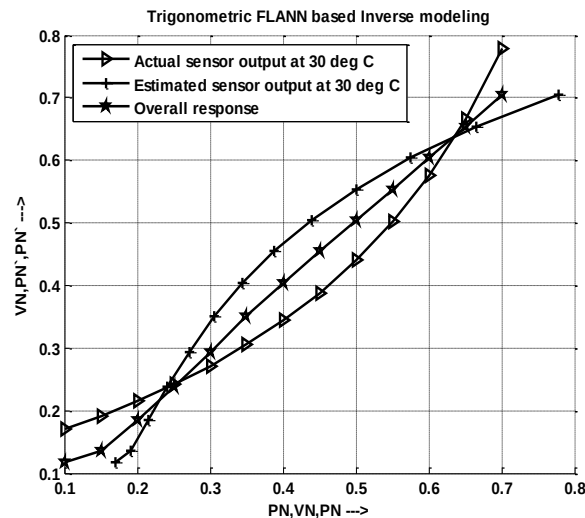
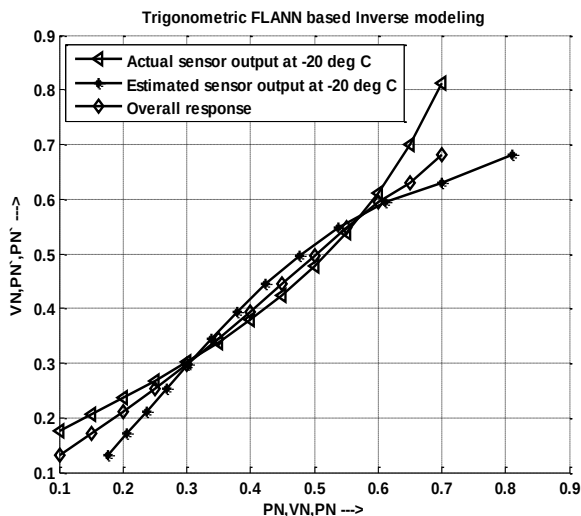


Simulation result of the trigonometric FLANN

No. of $\sin(\cdot)$ expansion = 10, No. of $\cos(\cdot)$ expansion = 10, (of each input)

Learning rate of the update algorithm = 0.035.

Plots are at $T = -20^{\circ}\text{C}$, 30°C and 100°C

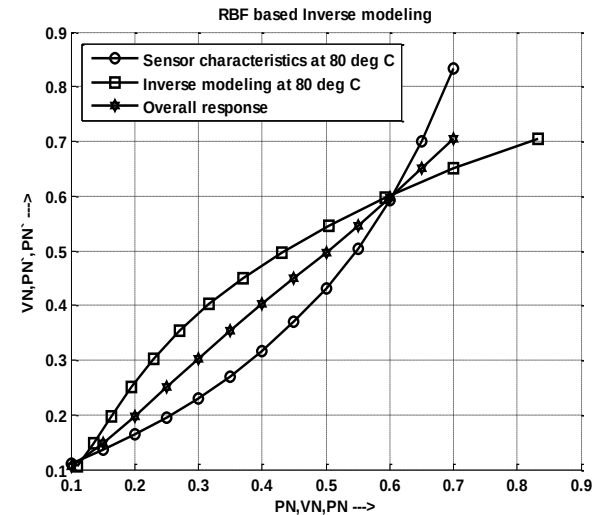
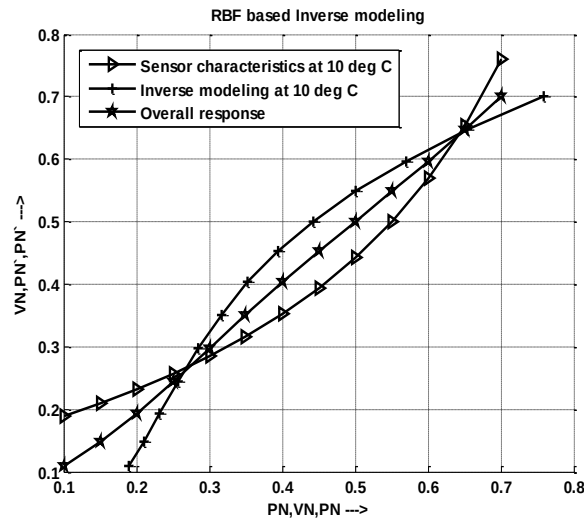
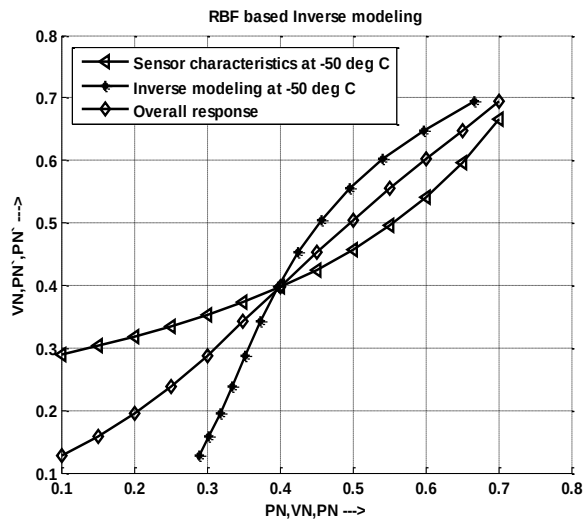


RBFNN based Inverse Modeling

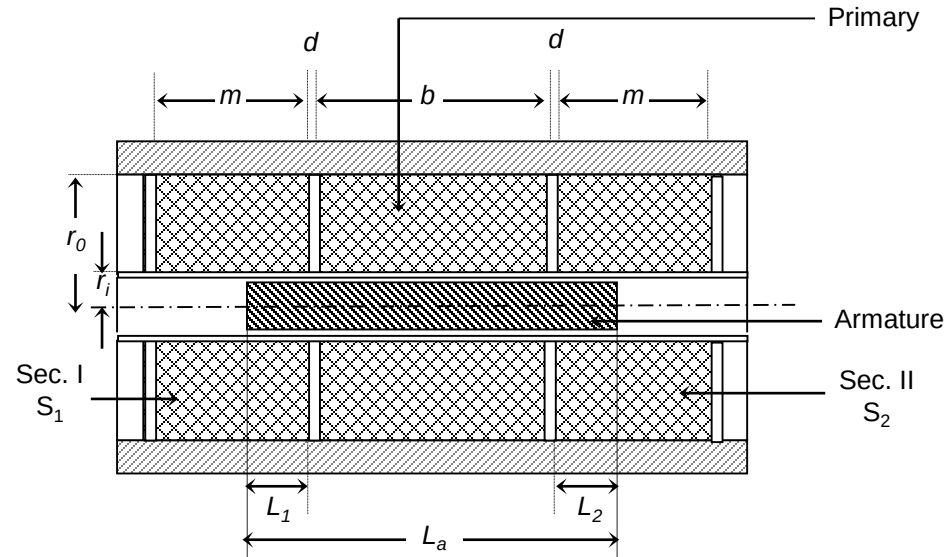
A 2-6-1 RBFNN is taken for simulation.

The learning rate of the updation algorithm is chosen as 0.05

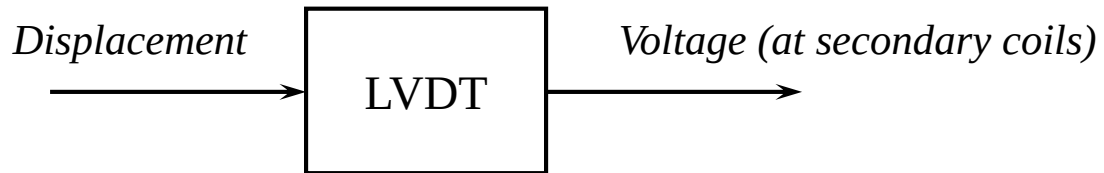
Figures are plotted at $T = -50^{\circ}\text{C}$, 10°C and 80°C .



Linear Variable Differential Transformer



Cross-sectional view of an LVDT



The differential voltage, $e = e_1 - e_2$

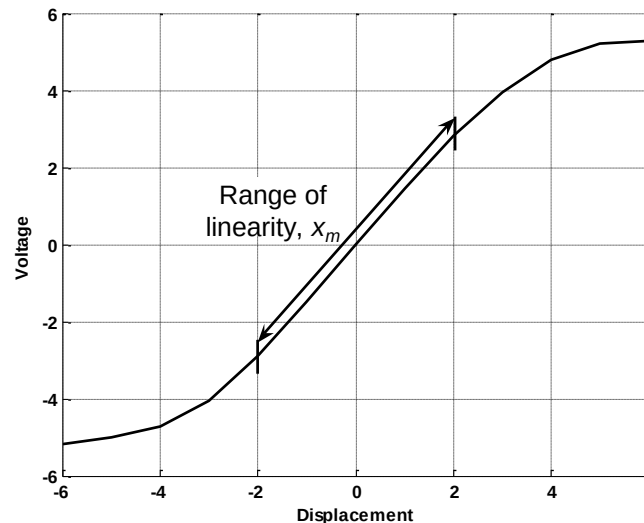
(e_1 = induced emf in secondary coil, S_1 and e_2 = induced emf at S_2)

$$e = k_1 x (1 - k_2 x^2)$$

Where,

k_1, k_2 = constants, (depends on the internal parameters of LVDT).

Input-output characteristics of a standard LVDT

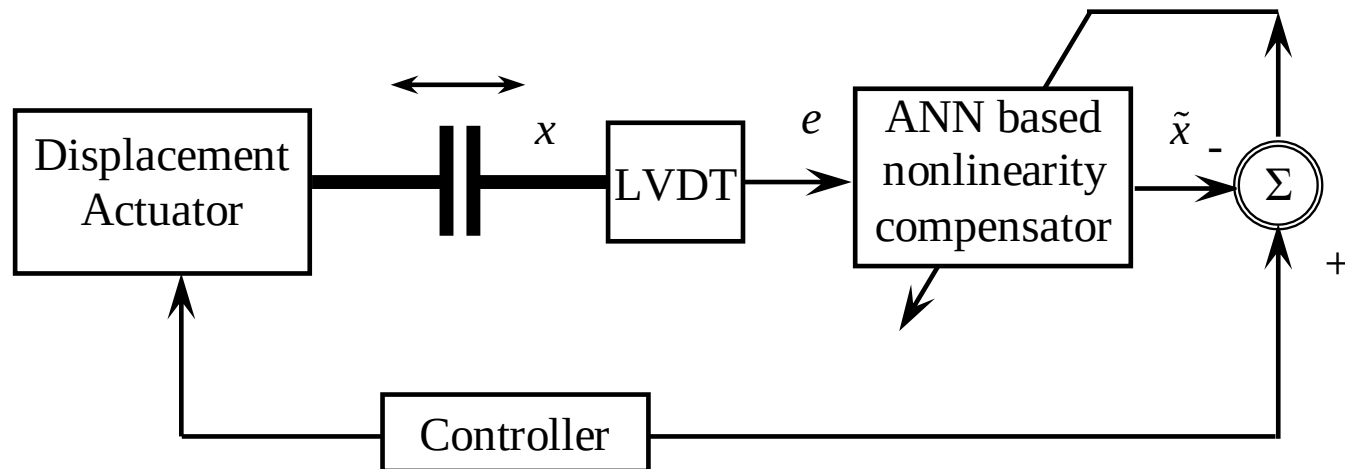


The range of linearity can be extended by appropriate design and arrangement of the primary and the secondary coils. Such as:

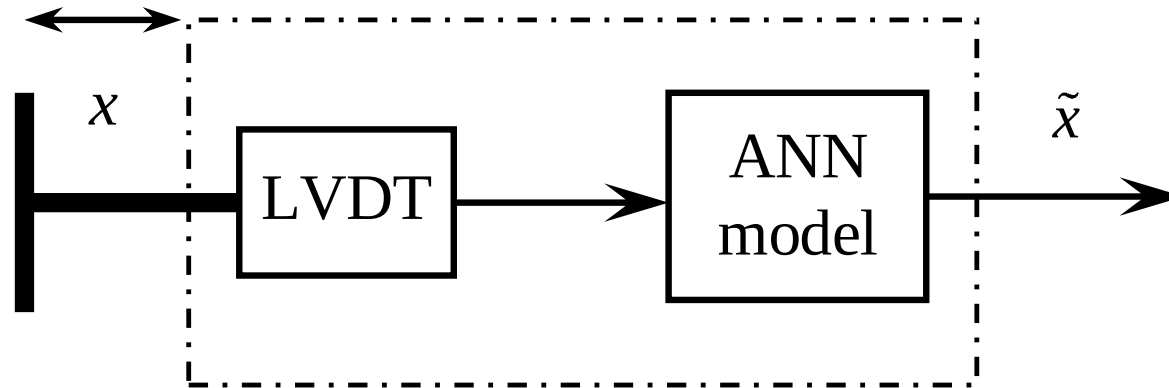
- Balanced linear tapered secondary coils. But the improvement in linearity is not significant.
- Over-wound linear tapered secondary coil improves the linearity to a certain extent.
- Balanced over-wound linear tapered secondary coils. Its performance is similar to (ii).
- Balanced profile secondary coils helps in extending linearity range by proper profiling of the secondary coils.
- Use of complementary tapered winding extends the linearity range but the winding is quite complicated as it involves sectionalised winding.

Nonlinearity Compensation of LVDT

The practical set-up of the LVDT along with the ANN based nonlinearity compensator is



Once the training is over,



The *ANN based adaptive nonlinear compensator* can be developed by

- MLP
- FLANN
- Cascaded FLANN (CFLANN)

Simulation Model

Specification of a standard LVDT

Two secondary coils (3300 turns) each wound over it uniformly with the two coils separated by a Teflon ring.

The core diameter = 4.4 mm, core length = 4.5 mm.

Excitation frequency, and voltage given to primary coil are 5 kHz and 10V (peak-to-peak), respectively.

$V_p = 6.712 V_{rms}$.

Primary winding resistance = 260 ohm.

Secondary winding resistance (S_1) = 426.8 ohm.

Secondary winding resistance (S_2) = 414 ohm.

Two secondaries are wound in opposite directions (one clockwise and the other anticlockwise). The pitch is kept around 0.02-0.03 mm.

The experimental measured data

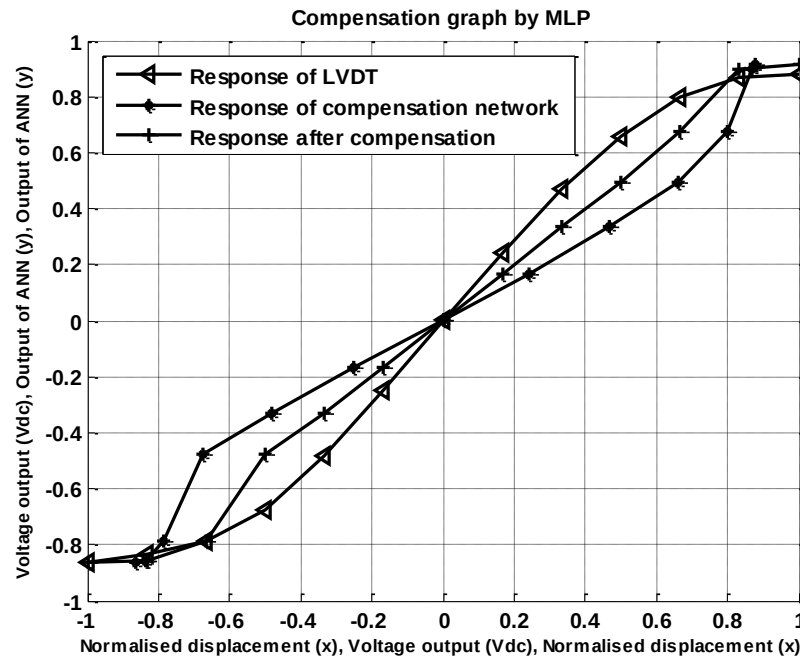
Displacement (x mm)	Output voltage (e_{rms})	Demodulated output (e_{DC})
-30	4.085	-5.185
-25	3.956	-5.017
-20	3.731	-4.717
-15	3.221	-4.039
-10	2.359	-2.896
-5	1.273	-1.494
Null Position, 0	0.204	0.001
5	1.153	1.462
10	2.226	1.810
15	3.118	3.962
20	3.748	4.799
25	4.050	5.225
30	4.085	5.276

MLP based Nonlinearity compensator

A 1-100-50-1 MLP network is taken for the simulation.

The two hidden layer and the output layer contain $\tanh(.)$ activation function.

The learning rate is taken as 0.02.

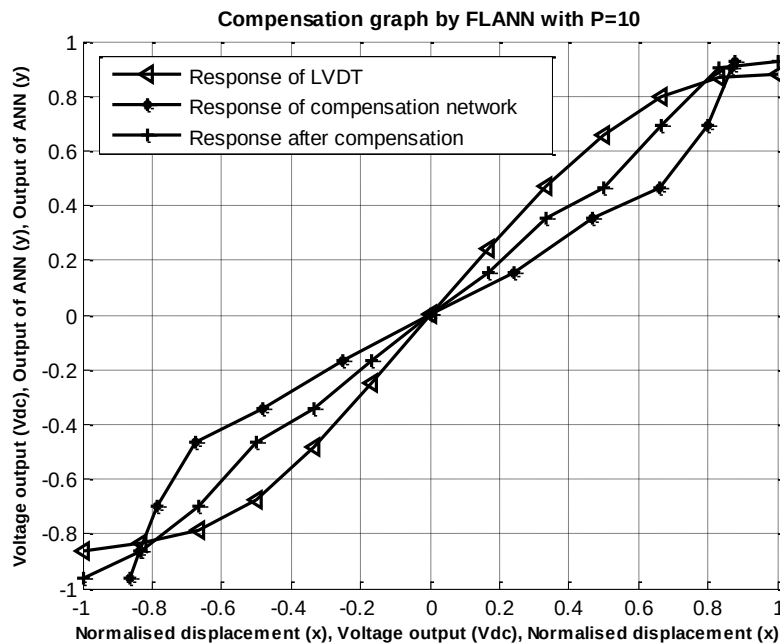


$$\text{Percentage of linearity, } PEL = \frac{(\text{desired output} - \text{estimated output})}{\text{desired output}} * 100 = 3.1120 \%$$

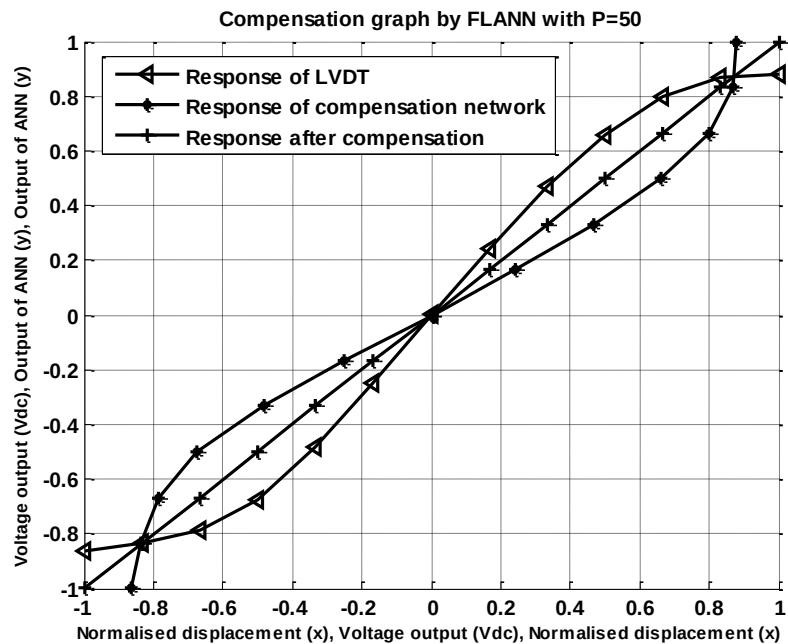
FLANN based Nonlinearity compensator

Two experiments of different expansion level has been conducted.

- 1) No. of expansion = 10 (5 sin(.) and 5 cos(.))
- 2) No. of expansion = 50 (25 sin(.) and 25 cos(.))



$$PEL = 0.3251 \%$$



$$PEL = 5.12 \times 10^{-15} \%$$

Thank You